

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 11 YEARLY EXAMINATION

MATHEMATICS EXTENSION 1

2006

Time allowed: 90 minutes

Directions to Candidates

- Attempt all questions
- Start each question on a new page
- All necessary working should be shown
- Unless otherwise specified, answers must be given in their simplest form
- Approved calculators may be used in all sections.
- Use a ruler when drawing straight lines
- Marks may be deducted for careless or poorly arranged work.
- Marks shown are approximate and may be varied.

Name: _____ Class: _____

1	2	3	4	5	6	TOTAL

Question 1

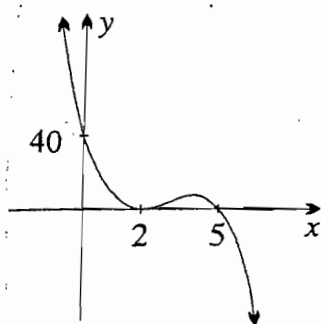
a) Fully factorise $2a^3 - 128$. (1)

b) i) Find the remainder when $P(x) = x^4 + 2x^2 - 5$ is divided by $(x-2)$.

ii) Explain why there is a zero in the domain $0 < x < 2$. (3)

c) If $4 + \sqrt{b} = \sqrt{19 + \sqrt{m}}$, find the values of b and m . (2)

d) This is the graph of $y = P(x)$. (3)



i) Write down the equation of $y = P(x)$.

ii) Write down the domain of $y = \sqrt{P(x)}$.

e) Find the gradient of the tangent to the curve $y = \sqrt{5 + x^2}$ at $x = -2$ as a fraction. (2)

Question 2 (Start on the next page)

a) Use long division to find the remainder when $x^4 - x^2 - x + 8$ is divided by $x^2 - 3$. (2)

b) If $f(x) = \frac{x^2}{x+4}$ (3)

i) Find $f'(x)$.

ii) Find the values of x if $f'(x) > 0$.

c) i) Explain why $|xy| = 4$ is not a function.

ii) Sketch the graph of $|xy| = 4$. (2)

Q2 (cont.)

- d) i) Write down the expansion for $\tan 2A$. (4)
- ii) Hence find the exact value of $\tan 22\frac{1}{2}^\circ$.

Question 3 (Start on the next page)

- a) Two roots of the cubic equation $x^3 + mx + n = 0$ are -3 and 4 . (2)
- i) Find the third root.
- ii) Find the value of n .
- b) The points $P\left(t, \frac{t^2}{2}\right)$ and $Q(-4, 8)$ are points on a parabola. (3)
- i) Find the cartesian equation of the parabola.
- ii) If PQ is a focal chord, what are the co-ordinates of P .
- c) If $\sec \theta - \tan \theta = x$, show that $x = \frac{1-t}{1+t}$ where $t = \tan \frac{\theta}{2}$. (3)
- d) i) Find the value of c if $P(x) = x^3 - 3x^2 - 4x + c$ is divisible by $x - 3$. (3)
- ii) Hence evaluate $\lim_{x \rightarrow 3} \frac{x^3 - 3x^2 - 4x + c}{x - 3}$.

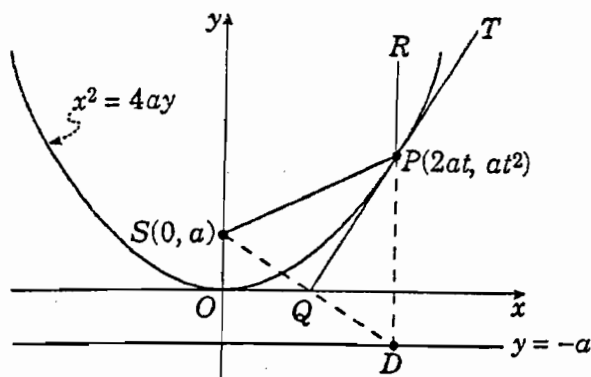
Question 4 (Start on the next page)

- a) i) Find the locus of the point P which is equidistant from the points $A(0, 2)$ and $B(6, 0)$. (5)
- ii) A point Q is closer to B than A and less than 4 units from A . Write inequalities which would describe the region where Q could be located.
- b) $T(6t, 3t^2)$ is a point on the parabola $x^2 = 12y$. The point D is at the intersection of the directrix and the y axis. (6)
- i) The point P divides TD internally in the ratio $2:1$. Write down the co-ordinates of P in terms of t .
- ii) Show that as T moves on the parabola $x^2 = 12y$ the locus of P is $x^2 = 4y + 8$.
- iii) Write down the focus and directrix of the locus of P .

Question 5 (Start on the next page)

a)

(8)



The diagram shows the parabola $x^2 = 4ay$ with focus $S(0, a)$ and directrix $y = -a$. The point $P(2at, at^2)$ is an arbitrary point on the parabola and the line RP is drawn parallel to the y axis, meeting the directrix at D . The tangent QPT to the parabola at P intersects SD at Q .

- i) Explain why $SP = PD$.
- ii) Derive the gradient m_1 of the tangent at P .
- iii) Find the gradient m_2 of the line SD .
- iv) Prove that PQ is perpendicular to SD .
- v) Prove that $\angle RPT = \angle SPQ$.

b) The sum of two roots of the equation $x^3 + kx^2 + mx + n = 0$ is zero.

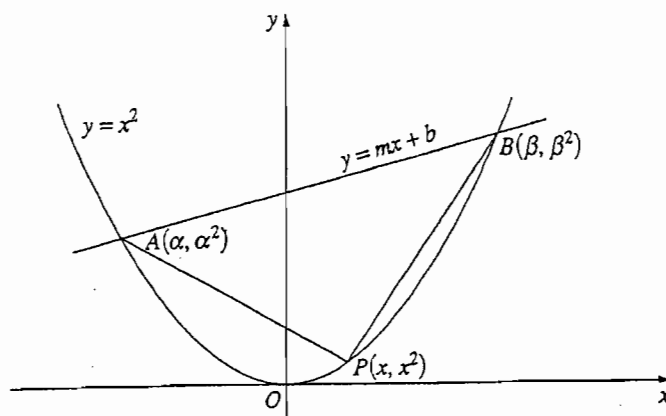
(3)

Show that $km = n$.

Question 6 (Start on the next page)

a) Solve $\cos 2x = \sin x$ for $0^\circ \leq x \leq 360^\circ$. (3)

b) (8)



The parabola $y = x^2$ and the line $y = mx + b$ intersect at the points $A(\alpha, \alpha^2)$ and $B(\beta, \beta^2)$ as shown in the diagram.

- Explain why $\alpha + \beta = m$ and $\alpha\beta = -b$.
- Factorise the expression $(\alpha - \beta)^2 + (\alpha^2 - \beta^2)^2$
- Hence or otherwise, using the fact that $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$, show that the distance AB is given by

$$AB = \sqrt{(m^2 + 4b)(1 + m^2)}$$

- The point $P(x, x^2)$ lies on the parabola between A and B. Show that the area of the triangle ABP is given by $\frac{1}{2}(mx - x^2 + b)\sqrt{m^2 + 4b}$.

- By treating $\frac{1}{2}(mx - x^2 + b)\sqrt{m^2 + 4b}$ as a quadratic expression in terms

of x , show that the maximum area of the triangle ABP is $\frac{(m^2 + 4b)^{\frac{3}{2}}}{8}$

End of Exam

①

$$a) 2a^3 - 128 = 2(a^3 - 64) \\ = 2(a-4)(a^2 + 4a + 16)$$

$$b) i) P(2) = 16 + 8 - 5 \\ = 19$$

$$ii) P(0) = -5$$

$P(x)$ changes sign between 0 and 2. $\therefore$ at least one zero

$$c) (4 + \sqrt{b})^2 = 19 + \sqrt{m} \\ 16 + 8\sqrt{b} + b = 19 + \sqrt{m}$$

$$\therefore b = 3$$

$$\text{and } 8\sqrt{3} = \sqrt{m}$$

$$\therefore m = 192$$

$$d) i) P(x) = -2(x-2)^2(x-5)$$

$$ii) x \leq 5$$

$$e) y' = \frac{1}{2} \cdot 2x \cdot (5+x^2)^{-1/2} \\ = x(5+x^2)^{-1/2}$$

$$\text{at } x = -2$$

$$y' = \frac{-2}{\sqrt{9}} \\ = -2/3$$

②

$$a) \begin{array}{r} x^2 + 2 \\ x^2 - 3 \overline{) x^4 - x^2 - x + 8} \\ \underline{x^4 - 3x^2} \\ 2x^2 - x \\ \underline{2x^2 - 6} \\ -x + 14 \end{array}$$

$\therefore$ remainder $-x + 14$

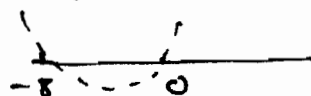
$$b) i) f(x) = \frac{x^2}{x+4}$$

$$f'(x) = \frac{(x+4) \cdot 2x - x^2 \cdot 1}{(x+4)^2} \\ = \frac{x^2 + 8x}{(x+4)^2}$$

$$ii) f'(x) > 0$$

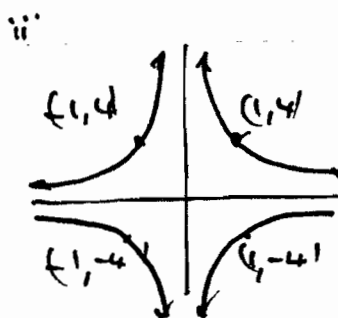
$$\frac{x^2 + 8x}{(x+4)^2} > 0$$

$$\frac{x(x+8)}{(x+4)^2} > 0$$



$$\therefore x < -8, x > 0$$

c) i) each value of x there is more than 1 value of y



$$a) \tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$ii) \text{ Let } x = \tan 22\frac{1}{2}$$

$$\therefore \tan 45^\circ = \frac{2x}{1-x^2}$$

$$1 = \frac{2x}{1-x^2}$$

$$1-x^2 = 2x$$

$$x^2 + 2x - 1 = 0$$

$$x = \frac{-2 \pm \sqrt{4+4}}{2}$$

$$= \frac{-2 \pm 2\sqrt{2}}{2}$$

$$= -1 \pm \sqrt{2}$$

$$\tan 22\frac{1}{2} > 0$$

$$\therefore \tan 22\frac{1}{2} = \sqrt{2} - 1$$

③ a) i) Let roots be
-3, 4, α

$$\therefore -3 + 4 + \alpha = 0$$

$$\alpha = -1$$

$$ii) -3 \times 4 \times -1 = -n$$

$$12 = -n$$

$$n = -12$$

b) i) $x = t$

$$y = \frac{t^2}{2}$$

$$\therefore y = \frac{x^2}{2}$$

ii) At Q $t = -4$

$$\therefore \text{ at P } t = 1/4$$

$$\therefore P(1/4, 1/32)$$

c) $\tan \theta/2 = t$

$$\sec \theta - \tan \theta = \frac{1+t^2}{1-t^2} - \frac{2t}{1-t^2}$$

$$= \frac{t^2 - 2t + 1}{1-t^2}$$

$$= \frac{(1-t)^2}{(1-t)(1+t)}$$

$$= \frac{1-t}{1+t}$$

$$\therefore x = \frac{1-t}{1+t}$$

d) i) For factor

$$P(x) = 0$$

$$\therefore P(x) = 27 - 27 - 12x + c$$

$$\therefore c = 12$$

ii) $\lim_{x \rightarrow 3} \frac{x^3 - 3x^2 - 4x + 12}{x-3}$

$$= \lim_{x \rightarrow 3} \frac{x^2(x-3) - 4(x-3)}{(x-3)}$$

$$= \lim_{x \rightarrow 3} \frac{(x^2 - 4)(x-3)}{(x-3)}$$

$$= \lim_{x \rightarrow 3} (x^2 - 4)$$

$$= 5$$

④ a) i) $PA^2 = PB^2$

$$\therefore x^2 + (y-2)^2 = (x-6)^2 + y^2$$

$$x^2 + y^2 - 4y + 4 = x^2 - 12x + 36 + y^2$$

$$12x - 4y - 32 = 0$$

$$3x - y - 8 = 0$$

ii)

Locus 4 units from A

$$x^2 + (y-2)^2 = 16$$

at B

$$\therefore 3x - y - 8 > 0$$

$\therefore$ region of intersection of

$$3x - y - 8 > 0 \text{ and } x^2 + (y-2)^2 < 16$$

b)

i) D (0, -3)

$$(6t, 3t^2) \quad P \quad (0, -3)$$

2 1

$$P = \frac{6t}{2}, \frac{3t^2 - 6}{2}$$

=

$$= (2t, t^2 - 2)$$

ii) $x = 2t$ ①

$y = t^2 - 2$ ②

From ① $t = \frac{x}{2}$

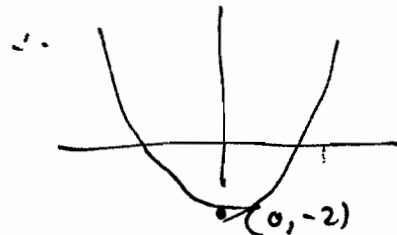
Sub in ②

$$y = \left(\frac{x}{2}\right)^2 - 2$$

$$4y = x^2 - 8$$

$$x^2 = 4y + 8$$

iii) $x^2 = 4(y+2)$



$\therefore$ focus (0, -1)
directrix $y = -3$

⑤ i) $SP = PD$ - definition of parabola

ii) $x = 2at$ $y = at^2$

$$\frac{dx}{dt} = 2a \quad \frac{dy}{dt} = 2at$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$= \frac{2at}{2a}$$

$$= t$$

iii) D = (2at, -a)

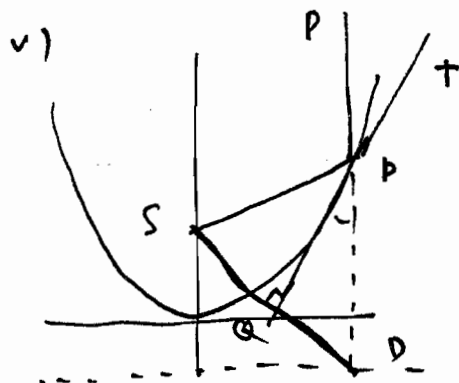
$$m_{SD} = \frac{a - (-a)}{0 - 2at}$$

$$= \frac{2a}{-2at}$$

$$\text{iv) } m_1 = k \\ m_2 = -1/k$$

$$\therefore m_1 \cdot m_2 = k \times -\frac{1}{k} \\ = -1$$

$$\therefore PQ \perp SO$$



$$\angle RPT = \angle QPD \\ (\text{vertically opposite } \angle)$$

$$\text{Now } \cos QPD = \frac{PQ}{PD}$$

$$\cos QPS = \frac{PQ}{SP}$$

$$\text{but } PD = SD \text{ proven in (i)}$$

$$\therefore \angle QPD = \angle QPS$$

$$\therefore \angle RPT = \angle QPS$$

$$\text{b) Let roots be } \alpha, -\alpha, \beta.$$

$$\therefore \beta = -K$$

but β is a root of the polynomial

$$\therefore P(-K) = 0$$

$$(-k)^3 + k(-k)^2 - mk + n = 0$$

$$\therefore mk = n$$

$$\text{⑥ } \cos 2x = \sin x$$

$$1 - \sin^2 x = \sin x$$

$$2\sin^2 x + \sin x - 1 = 0$$

$$(2\sin x - 1)(\sin x + 1) = 0$$

$$\therefore \sin x = 1/2, \sin x = -1$$

$$\therefore x = 30^\circ, 150^\circ, x = 270^\circ$$

$$\therefore x = 30^\circ, 150^\circ, 270^\circ$$

$$\text{b) i) } y = x^2$$

$$y = mx + b$$

$$\therefore x^2 = mx + b$$

$$x^2 - mx - b = 0$$

α and β are the roots of this equation.

$$\therefore \alpha + \beta = -\frac{b}{a} \\ = m$$

$$\alpha\beta = \frac{c}{a} \\ = -b$$

$$\text{ii) } (\alpha - \beta)^2 + (\alpha^2 - \beta^2)^2 \\ = (\alpha - \beta)^2 + [(\alpha - \beta)(\alpha + \beta)]^2 \\ = (\alpha - \beta)^2 [1 + (\alpha + \beta)^2]$$

$$\text{iii) } AB = \sqrt{(\alpha - \beta)^2 + (\alpha^2 - \beta^2)^2} \\ = \sqrt{(\alpha - \beta)^2 [1 + (\alpha + \beta)^2]}$$

from results

$$= \sqrt{[(\alpha + \beta)^2 - 4\alpha\beta][1 + (\alpha + \beta)^2]}$$

$$= \sqrt{(m^2 + 4b)(1 + m^2)}$$

iv)

Distance P to line AP

$$d = \frac{mx - x^2 + b}{\sqrt{1 + m^2}}$$

$$\therefore \text{Area} = \frac{1}{2} \cdot \sqrt{(m^2 + 4b)(1 + m^2)} \cdot \frac{mx - x^2 + b}{\sqrt{1 + m^2}}$$

$$= \frac{1}{2} (mx - x^2 + b) \sqrt{m^2 + 4b}$$

v) maximum value occurs when

$$x = -\frac{b}{2a}$$

$$= -\frac{m}{-2}$$

$$= \frac{m}{2}$$

$$\therefore \text{Area} = \frac{1}{2} \left[\frac{m^2}{2} - \frac{m^2}{4} + b \right] \sqrt{m^2 + 4b}$$

$$= \frac{1}{2} \left[\frac{m^2}{4} + b \right] \sqrt{m^2 + 4b}$$

$$= \frac{1}{8} [m^2 + 4b] \sqrt{m^2 + 4b}$$

$$= \frac{(m^2 + 4b)^{3/2}}{8}$$